

M.Tech Part I/PhD Mining Engineering, 2024-25
MN573: Safety & Risk Management in Mines

Time: 3 Hours

Max. Marks: 50

Note: Answer any five questions.

1. Explain the cost-of-illness approach and willingness-to-pay approach for accident cost evaluation. (10)
How can the concepts of 'value of statistical life' be utilised as a tool for accident prevention in Indian mines?
2. Explain various types of human errors. How do these human errors occur? Discuss with suitable examples, how these human errors can be eliminated from the industrial workplaces. (10)
3. Outline the objectives of the 'Safety Management Plan'? Explain the process of safety management plan. How can it be used as a tool for effective safety management in mines? (10)
4. What are the purposes of accident investigation in mines? What are various components of investigation? (10)
With respect to the procedure of accident investigation, elaborate in details the steps followed in 'examination of witnesses' and 'analysis of evidence'.
5. Define 'motivation'. How can motivation affect other behavioural traits of workers for injury occurrence/prevention in mines? (10)
Do you think Maslow's hierarchy of needs can address motivation as a human trait? — discuss. (10)
6. Write short notes on the following: (10)
 - (i) Risk-taking behaviour
 - (ii) Personality
 - (iii) Causal loop
 - (iv) Zero accident potential
 - (v) Random variable
7. (a). Calculate the expected value and standard deviation of Binomial random variable. (4)
8. (b). Outline limitations in application of Poisson's distribution. (2)
9. (c). Behaviour-based safety is believed to be a better alternative over the traditional safety practices. — discuss. (2)
10. (d). Safety and Risk are interrelated. — discuss. (2)
